

Applying Design and Agile Thinking to your Data Science project

Imagine the world as a gigantic puzzle with millions of pieces. The puzzle is still not complete, some pieces are already placed, but there is still space for new ones. We want to make our solution a puzzle piece that fits into this huge puzzle, but we do not know-how. **Design and Agile Thinking** help us shape our solution into a puzzle piece that fits into this huge puzzle.

Creating a data product can be chaotic and hard. My advice is to not give up easily when it at first seems

as if there is no solution, but to keep trying to build the pieces!

Following the **Design and Agile Thinking** as your guides in your trip can make the design, implementation, and arrival at your solution easier.

You have probably heard about the **Design and Agile Thinking** approaches since they are widely being used in many organizations. This tutorial gives a brief introduction to both approaches and explains how to apply **Design Thinking** in a **Data Science case**.

Design or Agile Thinking?

Difference #1

If you are in a situation that is blurry about what the actual problem is and what the tangible solutions are, then you can use the steps of the **Design Thinking** method to help your way through it.

If you are in a situation in which the problem is well-defined and you need to manage your way until you reach the solution on time, then you can follow the **Agile Thinking** method.

Difference #2

Design Thinking is used to redefine problems, understand users, experiment with novel concepts, and deliver optimal and innovative designs of solutions.

Agile Thinking is a project management method that transforms the design of a solution, which is the outcome of the Design Thinking process, to an actual solution.

Similarities

Both have similarities such as being:

- **human-centered**, setting as a priority the end user's and stakeholders' needs,
- **iterative**, consisting of multiple cycles and improving with the provided feedback in each iteration,
- **option-focused**, listing possibilities for exploration

The combination of those two methods can ensure high-quality solutions to real problems within organizations.

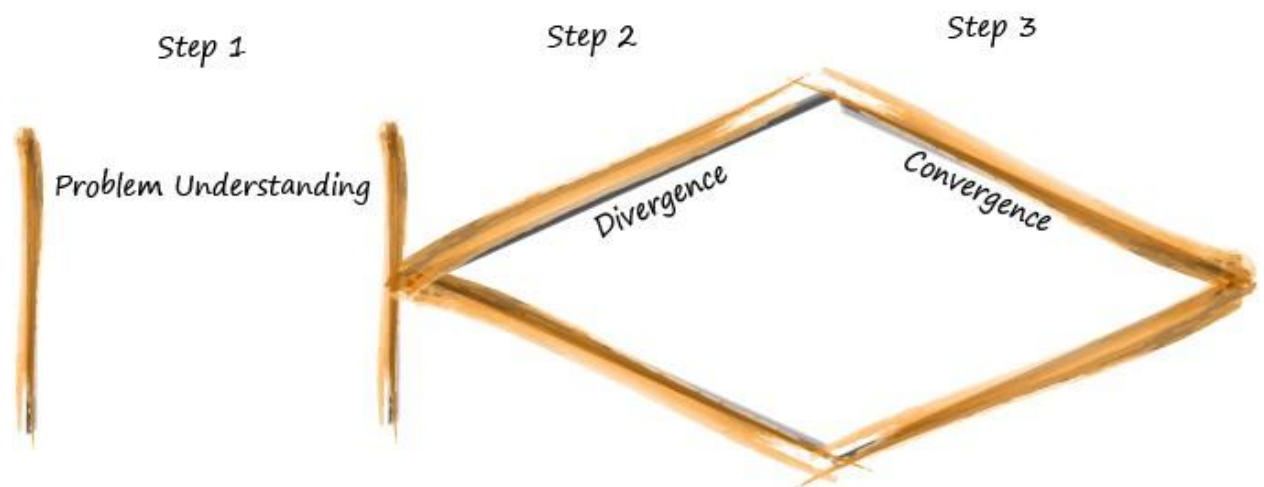
Design Thinking: Data science without data

In most cases, huge amounts of data are available and if we start immediately by looking at the data, we will probably get lost while trying to understand the data columns and fields and forget about the initial problem. So, we should take a step back, detach from the details and try to create a holistic overview of the organization's processes and its bottlenecks.

Our desired **outcome** in this phase is to **identify problems and design solutions**. Since we want

to bring a data-driven solution, the design should include an initial draft of the data requirements and a first concept of the data product. In this phase, we should also define the users of the end product and include them in the design process.

The Design Thinking approach consists of three steps; **problem understanding, divergence, and convergence**, as displayed in the diagram below.

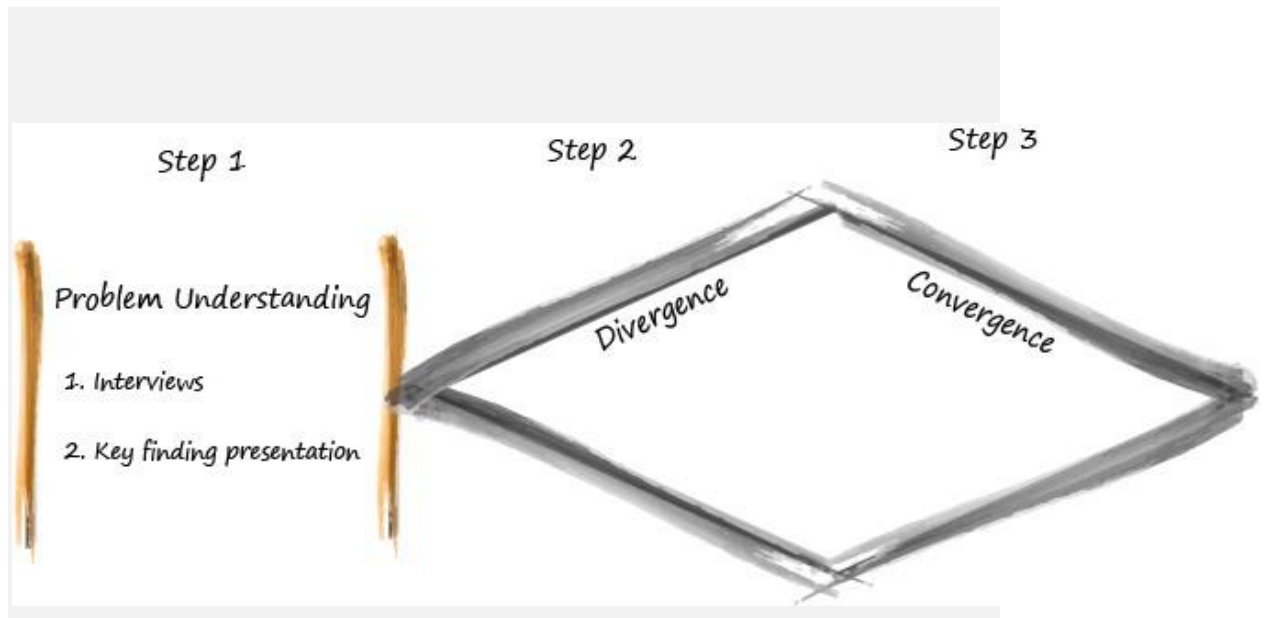


Design Thinking, Sketch by Elena Stamatelou

Step #1: Problem understanding

This phase aims mainly to a deeper

understanding of the current situation within the organization.



Design Thinking: Step 1, Sketch by Elena Stamatelou

A. Interviews

Arranging interviews with different people from the organization, helps us gain **insights** into their processes, communication styles, and vision. From a Data Scientist's perspective, the expected key

outcome is to discover the **data maturity level** of the organization, which shows the current role of Data Science in their organization. To reach this level of understanding, we made a list of questions for the interviewees.

Personal Questions

1. What is your background?
2. What is your role within the company? In which department, do you work?
3. How much do you know about Data Science?

Team Questions

1. Are you part of or do you manage a team?
2. What are the roles of the colleagues that you closely collaborate with?

3. What is the function of your team?
4. How is the team's structure in terms of size and hierarchy?
5. How frequently do you communicate within the team and with the other departments?

Business Questions

1. What are the most challenging problems of your organization?
2. What are the most challenging problems of your team?

Data Science Questions

1. Do you have any examples of Data Science related projects within your organization? If yes, what were those projects about?

2. Do you have any Data Scientists working for your organization? If yes, can you give some examples of their impact?
3. Do you have any idea or suggestion about how data science could handle challenges within your organization?

Data Pipeline Questions

1. Do you currently collect or possess any data?
2. Do you perform tasks that involve working or analyzing the data?
3. How is the data stored and extracted?
4. What is the size of the data?

Technologies

1. What type of dashboards do you use for data visualization/analytics?
2. What kind of database systems do you use for data storage?
3. Do you use any cloud technologies?
4. Do you have any preferred programming language for data analysis/modeling?

Vision Questions

1. Do you see your organization developing towards a more data-driven organization in the next ten years?
2. How easily do the employees adapt to new technologies, such as a new piece of software or data product?

B. Key findings presentation

The presentation should include a **summary of the outcomes** from the interviews and summarize the **major challenges** that the organization faces.

For adding the Data Science perspective, defining the **data maturity level** of the organization is crucial for moving the data use to the next level. The maturity can vary from a company not using any data to a company trusting the data and basing its decision-making on the data insights.

The following diagram shows the journey to a data-driven organization.



Data maturity level, sketch by Elena Stamatelou

1. **Data-denial:** active refusal of data usage
2. **Data-indifference:** lack of interest in the usage of the data
3. **Data-aware:** data collection and visualizations of historical data for past events understanding (descriptive analysis)
4. **Data-informed:** models based on the historical data for future event prediction (predictive analytics)

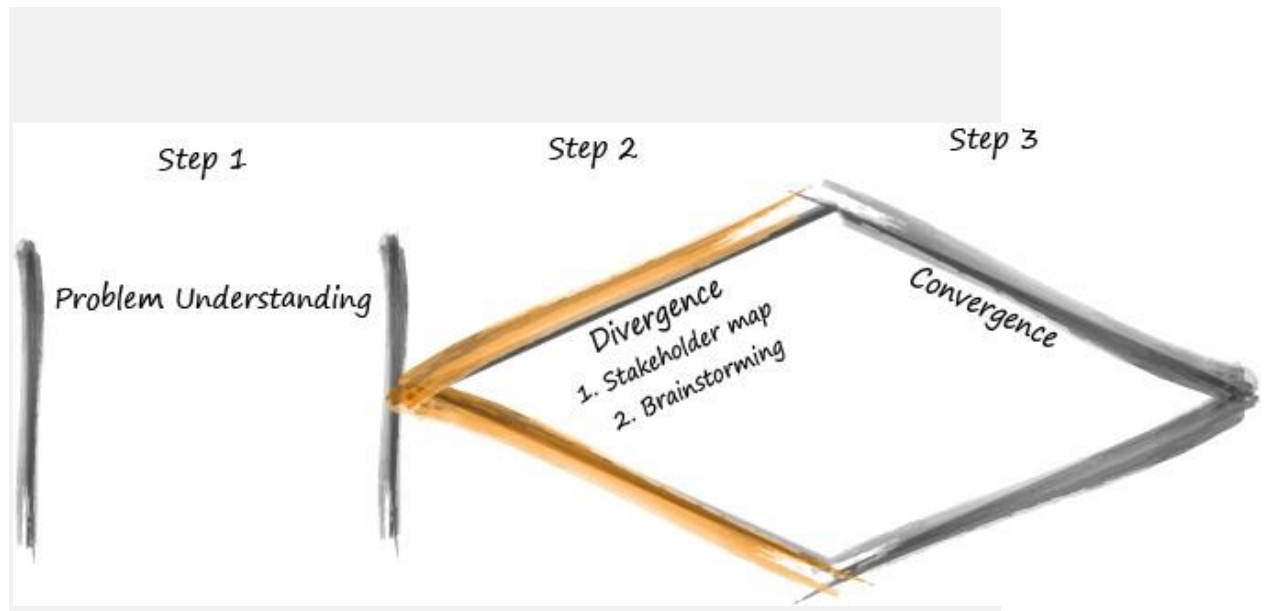
5. **Data-driven:** advice on the
decision-making based on the predictions
(prescriptive analytics)

After creating awareness about the key challenges and the organization's data maturity level, the presentation should also include some examples of how Data Science can help an organization handle some of its challenges and transform into a more data-driven one.

Step #2: Divergence

What if everything was possible? We should go through this phase with this question in mind and **explore every possibility**. The phase consists of

two parts; the first draft of the stakeholder map and the brainstorming session.

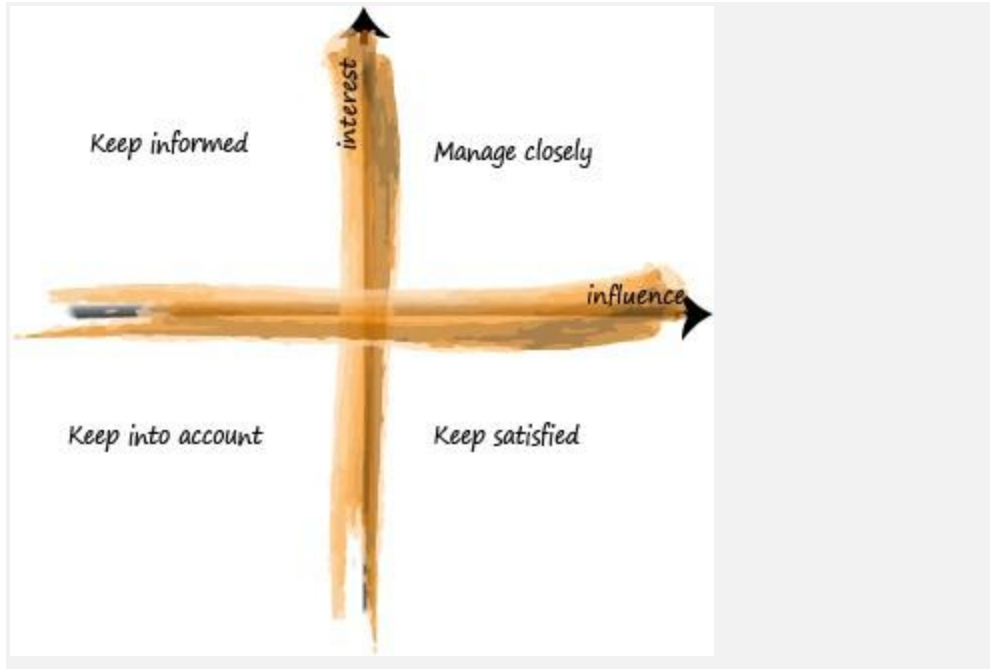


Design Thinking: Step 2, Sketch by Elena Stamatelou

A. Stakeholder map (first draft)

After the problem understanding phase, we gained a good overview of the parties involved in our project.

To organize this view of different stakeholders as well as their involvement and impact, we need to create the first draft of a stakeholder map, as displayed below.



Stakeholder map, sketch by Elena Stamatelou

The two main variables that differentiate the different types of stakeholders are **their interest and their influence level** in our project, and all combinations of those two variables are possible.

For now, we don't need to make a detailed map of all the stakeholders, but a first draft is enough to continue to the next steps of the Design Thinking method.

B. Brainstorming

Having completed the stakeholder map, we select the most representative stakeholders from the different parts of the map, so that we have a suitable combination of stakeholders with different impact and interest levels. Then we invite those stakeholders to our brainstorming session.

During the brainstorming, we create two or three groups with different stakeholders. We split the session into two parts; the ideation for challenges and the ideation for solutions.

For the **ideation for challenges** part, we ask the participants about the two major challenges they face, group them and let them vote for the most important ones. Then, for the most voted challenges,

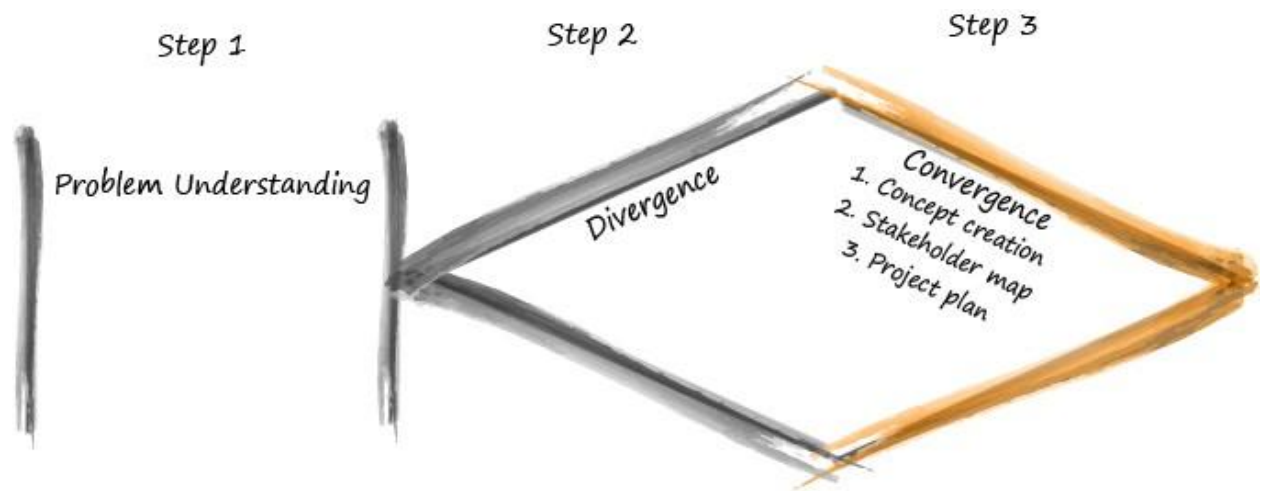
we request them to determine the stakeholders affected by them and group them together.

After that, we select the three most voted challenges and their most affected stakeholders and kick off the **ideation for solutions** part. Later, we ask the participants to imagine the ideal situation for the closely related stakeholders and to identify the bottlenecks that prevent the organization from reaching this ideal situation. Then, we let them brainstorm about ways for managing those bottlenecks and group them by similarity.

Step #3: Convergence

After diverging and exploring a variety of challenges and their possible solutions, adding no limitations, it is time to converge. The aim here is to define a

specific use case and propose a data-driven design concept to the stakeholders.



Design Thinking: Step 3, Sketch by Elena Stamatelou

A. Concept creation:

Our priority here is to summarize all the challenges and solution ideas that were created during the brainstorming session and focus on the ones that seemed to have the greatest impact on the organization and to be more burdensome. For each one, we work on defining a first draft of the data

requirements, a concept of the data product, and a high-level data-driven data pipeline.

For the first draft of the **data requirements**, we think about data availability, possible data collection, and data privacy issues. After that, we work on making a first sketch of the **final data product**, incorporating the initial data requirements. With the data requirements and the end product in mind, we design an overview of a **data pipeline** that includes the entire process from the data collection to the deployment of the final product and its delivery to the stakeholders.

Having a good first design of the solutions of the key challenges defined during the brainstorming, we arrange meetings with the main stakeholders. We

use the time of these meetings to discuss those designs and converge to one design that solves a use case that seems to be more interesting for them and more realistic to be solved.

B. Stakeholder map refinement

After having defined a first design of the solution and a specific use case, it is time to review the first draft of the stakeholder map, which was created before the brainstorming session.

The primary purpose of this step is to refine the different groups of stakeholders and create three teams; the **data, feedback, and management team**. Those teams are essential for the co-creation process of the development of the product and for enabling effective collaboration with them.



Teams, sketch by Elena Stamatelou

The **data team** will include stakeholders that are closely related to the data such as Data or Business Analysts. This team will help with data collection and data privacy. The **feedback team** will consist of the end-users that will experiment with the prototype and the end product and provide their feedback. In the **management team**, the stakeholders higher in the hierarchy will be included. This team will serve as the steering committee and will have a high impact on the decision-making of the project direction.

C. Project plan

The last step before working towards implementing the design is to create a project management plan.

This is a transition step between the Design Thinking and the Agile Thinking method.

It is essential to **review** all the information created during the Design Thinking process, including the data requirements, the design of the solution, and the different groups of stakeholders. Based on this review, we create an agile plan, after its execution will deliver the data product.

Agile Thinking: Data science with data

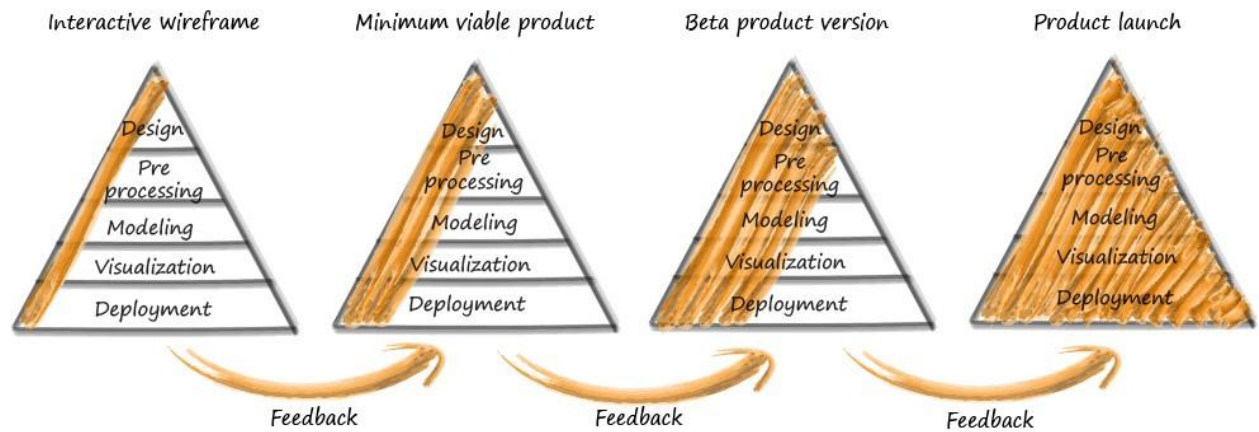
Given the design of the solution, which was the outcome of the Design Thinking, the Agile Thinking

method facilitates the implementation process of the design and ensures the delivery of the final data product in time. While during the Design Thinking the use of data was not necessary, in the Agile Thinking phase the use of data is required, but not necessarily from the beginning. Our desired **outcome** at this point is to launch the data produced within a timeframe.

Agile Thinking is an iterative approach, consisting of phases. Each phase will gradually build on top of the previous one with the product launch to be the eventual goal. We start with a very basic outline of our product, which is the **interactive wireframe** (phase 1), and we continuously add functionalities to enrich each time the latest version. From the basic first version, we move swiftly and release a

minimum viable product (phase 2) and then from that, the **beta product version** (phase 3). Finally, we deliver our final data product during the **product launch** (phase 4).

Each phase consists of the same five components; **design, preprocessing, modeling, visualization, and deployment**. The phases together with their basic components are displayed as pyramids in the diagram below. Every phase considers the same aspects, but each time builds on top of them by improving them and making them more complete.



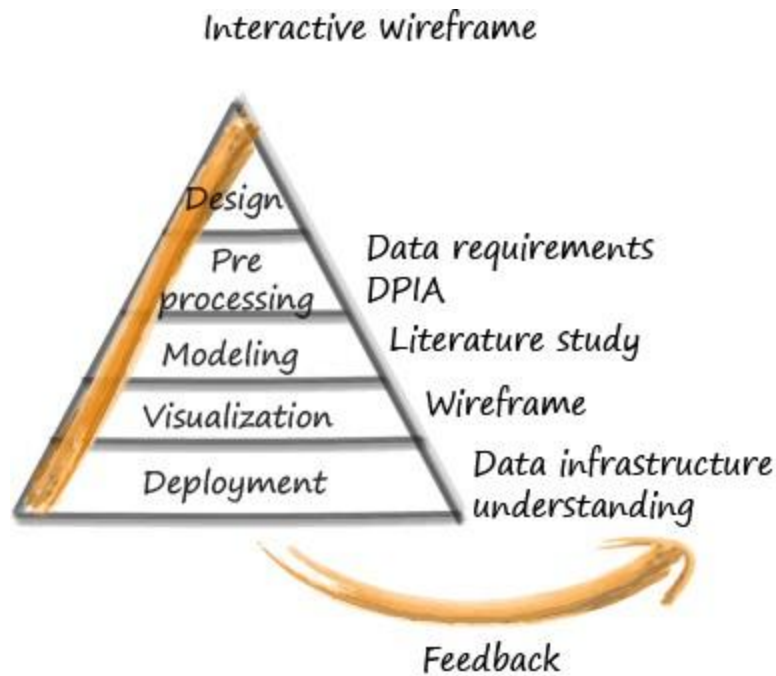
Agile Thinking life-cycle, sketch by Elena Stamatelou

At the beginning of each phase, there is some space for redefining our solution approach, by considering the learnings from the previous phase. This gives the possibility to combine the flexibility for change, as well as attachment to our initial direction, which was the outcome of the Design Thinking method. At the end of each phase, **feedback** sessions are necessary to be planned with the end-users of the data product (feedback team). That feedback provides knowledge about their needs and their level of satisfaction regarding the current version of the

product. As a result, that helps to define the necessary modifications that need to be applied in the next phase.

Phase 1: Interactive wireframe

This phase covers the project kick-off, having no data yet. The main deliverable is the **product outline** and an **interactive wireframe** which will be a very basic user interface and will enable the users to get a first impression of our direction.



Agile Thinking: Interactive wireframe (phase 1), sketch by Elena Stamatelou

Design

We use the outcome of the Design Thinking method to define the **initial architecture of the solution**, the tasks, and the direction of this phase.

We also create an initial sketch of the user stories, which we will use as a guide during the phase.

Preprocessing

Data requirements: Since there is no data, the primary aim here is to define the data requirements. We consider two opposite directions. On one side, by imagining that there are no restrictions regarding the data availability, we define the necessary data to solve the challenge. From the other side, by considering the existing data and its restrictions, we think of ways to achieve the final project goal. Then, we end up with a list with all the required data.

Data Protection Impact Assessment (DPIA):

Before getting the data, we need to prepare and sign an agreement with the organization to ensure data protection. We work on the DPIA process to identify and minimize possible data protection risks.

Modeling

Literature study: Before starting with any modeling approach, we need to make a literature review. Our aim is to create a list with all the possible modeling methodologies, from the best practices to the state-of-the-art.

Visualization

Wireframe: Having in mind the user perspective and needs that are described in the user stories, we create the format and the general structure of the visualizations. We also make some high-level sketches and we combine them all into an interactive wireframe, which will be used for the feedback session at the end of the phase.

Deployment

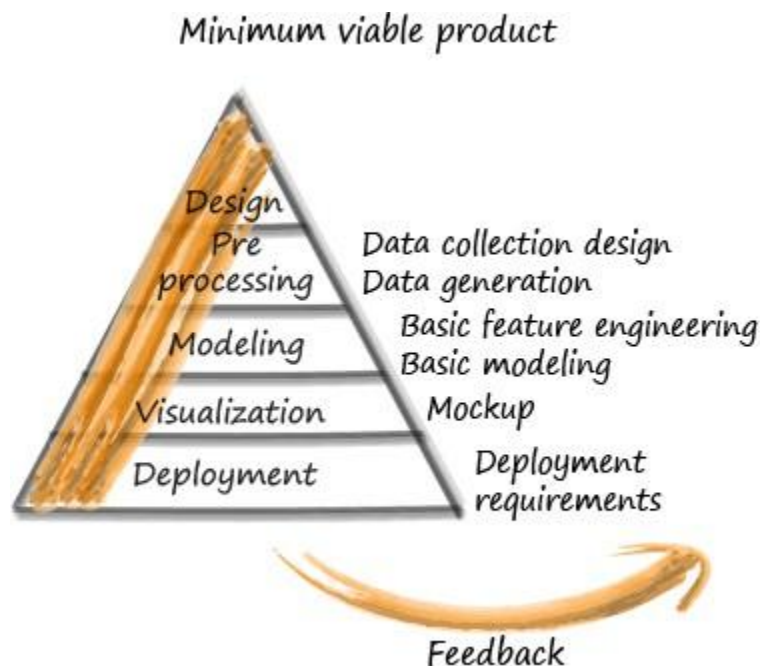
Data infrastructure understanding: We gather information about the organizations' infrastructure, such as the data storage or BI tools. Our goal is to figure out which tools we will use for the deployment of our data product. It is more cost-efficient and easy to use the tools that the employees are familiar with than trying to introduce new ones.

Feedback

At the end of the phase, we organize a feedback session, in which we present to the users the interactive wireframe, let them use it, and ask them questions to get their feedback.

Phase 2: Minimum viable product

Our goal, here, is to provide a foundation of the solution to the stakeholders and end-users, which will be the **minimum viable product**. From our side, we want to collect the maximum amount of feedback and learnings, so that we can ensure that our final working prototype will satisfy user's and stakeholder's expectations and needs.



Agile Thinking: Minimum viable product (phase 2), sketch by Elena Stamatelou

Design

We use the feedback collected in the previous phase, to refine the user stories, reshape our design, and plan the current phase. We **update the initial architecture** by adding more details.

Preprocessing

Data collection design: Having the list with the data requirements, we design a way to get the data.

If the data already exists in their systems, it is easy to request it and receive it. If the data does not exist yet, we should define a data collection process.

Data generation: Since it takes some time for the data collection, to save time, we generate synthetic data, based on the data requirements. For this phase, we use the generated data.

Modeling

Basic feature engineering: Before having the real data, we use the synthetic generated data to create basic features that we think are the most influential. We also clearly define the target variables.

Basic modeling: Using the basic features, we apply simple models to predict our target variables.

Visualization

Mockup: Using as a starting point the wireframe of the previous phase, we add data visualizations based on the updated user stories. Our focus is to make a user-friendly and interactive mockup that will make the users sense our direction and provide us constructive feedback.

Deployment

Deployment requirements: Based on the knowledge of the organization's infrastructure, we specify the deployment requirements and check the feasibility and the costs involved.

Local deployment: For this phase, we run the model in our local server,

Feedback

Before the phase ends, we arrange a meeting with the users to share with them our concept and the data visualizations of the predictions by showing them our minimum viable product.

Phase 3: Beta product version

This phase focuses on delivering a **beta product version**, which includes the real model predictions and displays them in a user-friendly and interactive dashboard. At the end of this phase, users and stakeholders will test the product to identify bugs and other deficiencies, in order to go to the product delivery.



Agile Thinking: Beta product version (phase 3), sketch by Elena Stamatelou

Design

Before starting the phase, we reconsider all the learnings and update the architecture by adding details from the data storage to the model deployment. This phase is one step closer to the actual product since we will work with real data.

Preprocessing

Data collection: Based on the data collection design and data requirements, we gather the data from the organization.

Data exploration: Having received the real data, we check the completeness (missing values) and timeliness (up-to-date data) to ensure data quality so that we can use it further for modeling. To gain a better understanding of the unexpected behaviors of the data, we look for outlier values (anomalies)

which are false values or very important ones, showing a strange behavior of the data.

Modeling

Feature engineering: To continue using the same or similar modeling approach, we examine the extent to which the provided data can replace the synthetic data. Then, we match the features of the synthetic data with the features of the real data and add more features from the real data if we think they will improve the prediction results.

Advanced modeling: Having the features from the real data, we first apply the simple model from the previous phase to ensure that the features with real data can replace the synthetic ones. If the simple model is not suitable anymore, we will

replace it with a more suitable or more complex one.

Here, we could also experiment with different modeling approaches or with hyperparameter tuning and compare the performance of the different models or variations of the same model.

Visualization

Prototype: Moving from the interactive mockup of the previous phase to a prototype, requires working on further improvement of the data visualizations and on the front-end development of the dashboard.

The focus here is to make a functional and interactive dashboard that can serve the end-user.

Deployment

Cloud/in-house deployment: Based on the deployment requirements that were defined in the

previous phase, we should move from having the solution running locally to our devices to running it in the organization's cloud or in-house servers. In this part, we deploy the prediction model and not the training. If everything runs smoothly, in the next phase, we will also deploy the training model.

Feedback

This phase results in a Beta product version that resembles the actual product since we worked both on the front-end (dashboard) and the back-end development (deployment). At this point, we give the last chance to get feedback from the users before the product launch.

Phase 4: Product Launch

In this phase, we apply the final improvements.

When no improvements are needed, we deploy the training and prediction model by making continuous validation possible and then launch the data product. Our goal here is to deliver a functional, reliable, and usable product to the stakeholders and users.



Agile Thinking: Product launch (phase 4), sketch by Elena Stamatelou

Design

In this last design cycle, we gather the feedback from the last feedback session and add the last refinements to the product architecture before the release.

Preprocessing

Preprocessing validation: We collect new data from the organization and compare it with the data used for the beta product version to detect differences.

Sanity checks: Based on the observation from the validation of the preprocessing with the new data, we create sanity checks to ensure that the input data has the suitable and desired format.

Modeling

Modeling validation: With the new data, we validate the beta product version and apply the last changes if needed.

Visualization

Dashboard: From the prototype of the previous phase, we build a more stable dashboard, in which we apply the last refinements.

Deployment

Continuous deployment: If the prediction model of the beta product version runs smoothly in the cloud or in-house infrastructure, we can go one step further to the deployment of our model. First, we also deploy our training model so that it reruns in case of a code update or new data. Our aim here is to achieve continuous integration (CI), continuous

delivery (CD), and continuous training (CT) for our prediction model. This means that with any new data or code update, the model is retrained and delivers the updated predictions.

Wrapping up

Finally, our product is ready to be launched!

With continuous integration, development, and training, our prediction model improves with new input. Providing this flexibility makes our data product adaptable to changes. By exposing it to new and more data improves the predictions of the data product.

Agile Thinking created a guideline to create a data product by taking as an input the design concept from the Design Thinking.